

Safety Manager Module Configuration Forms

**for use with the Triconex TRICON
Version 8 and Version 9 Systems**

SM88-600

**Implementation
Safety Manager**

***Safety Manager Module
Configuration Forms***
**for use with the Triconex TRICON
Version 8 and Version 9 Systems**

**SM88-600
06/06**

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About This Publication

This publication provides the configuration forms for the Safety Manager in TPS Release 670 series for the Triconex TRICON Version 8 and Version 9 Systems. These forms should be used to assist you in configuring the SMM data points at the Universal Station or the Global User Station.

The user should be familiar with the Safety Manager system control functions described in *SMM Control Functions for the Triconex TRICON Version 8 and Version 9 System*, and the *Implementation/Startup & Reconfiguration - 2* binder (LCN) bookset before using this publication.

This publication supports TPS system network through Release 670. TPS is the evolution of TDC 3000^x.

All references in this manual to “Safety Manager” or “Safety Manager Module” pertain only for use with the Triconex TRICON Version 8 and Version 9 Systems unless otherwise noted.

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Safety Manager Form Instructions

FILLING OUT PAPER FORMS

Where To Find Parameter and Data Point Help Information

Refer to the *Safety Manager Module Parameter Reference Dictionary* and to the *Safety Manager Module Control Functions* publications in the *Implementation Safety Manager* binder.

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SM Analog Input Data Point

SM Analog Input Data Point	Engineer:	Date:
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NAME
Tag Name

PNTFORM
Point Form

PTDESC
Point Descriptor

EUDESC
Engineering Units Descriptor

KEYWORD
Point Keyword Descriptor

PVSRCOPT
PV Source Option

PVSOURCE
PV Source

PVHITP
PV High Alarm Trip Point

PVHIPR
PV High Alarm Priority

PVLOTP
PV Low Alarm Trip Point

PVLOPR
PV Low Alarm Priority

PVHHTP
PV Hi Hi Alarm Trip Point

PVHHPR
PV High High Alarm Priority

PVLLTP
PV Low Low Alarm Trip Point

Point Assignment Display

range = Not applicable
default = Blank

Point is fully displayed and alarmed
Pt. is partially displayed but not alarmed

N/A if PNTFORM = Component.

range = Not applicable
default = Blank

range = Not applicable
default = Blank

range = Not applicable
default = Blank

N/A if PNTFORM = Component.

OnlyAuto
All

Auto
Man
Sub

PV source selection is not available and field wiring or memory fetch supplies PV. Selection from PVSOURCE is available & Operator/CL-program may supply PV.

Applies when PVSRCOPT = All.
Field wiring or memory fetch supplies PV. PV is supplied by Operator or program. A value is substituted by a CL program.

Alarming Display

N/A if PNTFORM = Component.

range = PVLOTP - PVHITP, & NaN
default = NaN

Noaction Journal Low High Emergency

range = PVLLTP - PVHITP, & NaN
default = NaN

Noaction Journal Low High Emergency

range = PVHITP - PVEUHI, & NaN
default = NaN

Noaction Journal Low High Emergency

range = PVEULO - PVLOTP, & NaN
default = NaN

Universal Station's configuration display where values and parameter options that have been written on the form will be keyed in and/or selected respectively.

If parameters within a box do not apply when some other parameter option is selected, that box is indicated by **N/A** if or **ONLY** if.

Text in **bold** indicates what parameter/s or parameter-option/s apply when an option is selected for a previous parameter.

Lower-level alarm trip points must have a value specified before higher-level alarm trip points display. For example, the PVHHTP (PV Hi Hi Alarm) parameter line displays after a value is entered in the PVHITP (PV High Alarm) parameter.

FORMS

ENTERING DATA FROM THE FORMS INTO THE SYSTEM

Select **NETWORK INTERFACE MODULE** from Engr's Main Menu to get NIM BUILD TYPE SELECT MENU.

UCN NODE CONFIGURATION Form SM88-601A	LM NODE SPECIFIC CONFIGURATION Form SM88-601B	PROCESS POINT BUILDING Forms SM88-602-610
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Select a point type from this POINT BUILT SELECT MENU

Analg In/Out	Dig In/Out	DigComp	Safety	Numeric	Flag	Timer
--------------	------------	---------	--------	---------	------	-------

SM UCN Node Configuration

SM UCN Node Configuration	Engineer:	Date:
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UCN Node Configuration Display

NOTE

This form is intended primarily for use in configuring each SM on your system by selecting **SM** for **NODETYP** below. This same form can be used to configure each NIM on your system by selecting **NIM** for **NODETYP** (some different targets will appear as shown). If your system has other UCN devices (PM, APM, HPM, or LM), the NIMs can also be configured using the forms contained in the documentation for those devices.

NTWKNUM NIM's UCN that SM is connected to (Process Network Number)	__.	range = 1 to 20 default = Not applicable
NODENUM SMM's or NIM's Address on the UCN	__.	range = 1 to 64 default = Not applicable
NODETYP Type of Node on the UCN	NotConfig Nim PM APM HPM LM SM	This NIM or SM not configured on UCN Network Interface Module Process Manager Module Advanced Process Manager Module High Performance Process Manager Module Logic Manager Module Safety Manager Module
SMPLTFM Type of Safety Control Platform	TRICON FSC	Safety Manager is using TRICON Controller Safety Manager is using FSC Controller
NODEASSN Node Assignment	ThisNim RemotNim	NIM assigned to report alarms and perform checkpointing for network specified in NTWKNUM Additional NIM on a network that already has "ThisNIM" defined
LOADSCOP Load Scope	NimOnly NimAndPm	Applicable only if NODETYP = Nim Configured data goes to the NIM only Configured data goes to NIM and SM
NMSGTXT Number of Message Text Items	__.	range = 0 to 15 default = 0
TIMESYNC SOE Time Synchronization	Disable Enable	Sequence of events times synchronization for this network

DIGITAL COMPOSITE PV STATE TEXT

MOVPTXT Moving Text	Moving	Individual point configuration specifies which input state is associated with this moving text. If NODENUM is even, this parameter is hidden.
BADPTXT Bad Text	Bad	Individual point configuration specifies which input state is associated with the bad text. If NODENUM is even, this parameter is hidden.

SM UCN Node Configuration

SM UCN Node Configuration	Engineer:	Date:
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Status Message Text Configuration Display

N/A if NMSGTXT = 0

MSGTXT(1) Status Message Text #1	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(2) Status Message Text #2	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(3) Status Message Text #3	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(4) Status Message Text #4	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(5) Status Message Text #5	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(6) Status Message Text #6	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(7) Status Message Text #7	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(8) Status Message Text #8	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(9) Status Message Text #9	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(10) Status Message Text #10	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(11) Status Message Text #11	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(12) Status Message Text #12	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(13) Status Message Text #13	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(14) Status Message Text #14	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank
MSGTXT(15) Status Message Text #15	_ _ _ _ _ _ _	<i>range =</i> Eight Character String <i>default =</i> Blank

***The Range check for this Parameter will be dependent on NNUMERIC value. SMM will perform Range checking.

SM Node Specific Configuration

NFLAG

|_|_|_|_|

range = 0 to 2000

of Flags

default = 0

The Range check for this Parameter will be dependent on the FLLSBA. SMM will perform Range checking.

FLLSBA

|_|_|_|_|

range = 0, 2001 to 4000, or 10001 to 14000

Starting Alias Address of the Flag Array

default = 0 (Not Configured)

The Range check for this Parameter will be dependent on the NFLAG value. SMM will perform Range checking.

NTIMER

|_|_|_|_|

range = 0 to 1500

of Timers

default = 0

The actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired. The 0.5 second scan rates will tend to lower the maximum range value. SMM will perform Range checking.

SCANRATE

SMM's Scan Rate in Times per Second

AR1DT1

AR1DT2

AR2DT2

[illegible]

SM Analog Input Data Point

NIM/PV Configuration Display

PVCHAR PV Characterization	LINEAR SQROOT TC_RTD
--------------------------------------	---

N/A if PVCHAR = TC_RTD				
PVRAW HI	_ _ _ _ _ _ _	default = 1.0	range =	≥ PVRAW LO
PVRAW LO	_ _ _ _ _ _ _	default = 0.0	range =	< PVRAW HI

N/A if PVCHAR = Linear, SqRoot		
PVTEMP Temperature Option	DegreesF DegreesC	F = Fahrenheit C = Celsius

Only if PVCHAR = Linear, SqRoot				
INPTDIR Analog Input Direction	Direct Reverse		Highest energy from sensor = 100% PV Highest energy from sensor = 0% PV	
PVEUHI PV High Range in EUs	_ _ _ _ _ _ _		<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0	
PVEULO PV Low Range in EUs	_ _ _ _ _ _ _		<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0	
PVFLTOPT Enumeration of the TRICON data type	_		<i>range:</i> 1 = Integer 2 = Real	
The value for this Parameter will be determined by PLCADDR. <i>default</i> = Not Configured				
PVFORMAT PV Decimal Point Format	D0 D1	D2 D3	D0 = 9999 o D1 = 999 o 9	D2 = 99 o 99 D3 = 9 o 999
PVEXEUHI PV Range Extension High	_ _ _ _ _ _ _		<i>range</i> = Depends on characterization <i>dault</i> = 102.9	
PVEXEULO PV Range Extension Low	_ _ _ _ _ _ _		<i>range</i> = Depends on characterization <i>default</i> = -2.9	
PVCLAMP PV Clamping Option	NoClamp Clamp		No clamping of the PV value Clamp PV value at range extension limit	

Only if PVCHAR = Linear, SqRoot		
LOCUTOFF Low Signal Cutoff for Flow Inputs	_ _ _ _ _ _ _	range = PVEULO to PVEUHI and NaN default = NaN

N/A if PNTFORM = Component		
PVSRCOPT PV Source Option	OnlyAuto All	PV source selection is not available and field wiring or memory fetch supplies PV. Selection from PVSOURCE is available and Operator/CL-program may supply PV.

SM Analog Input Data Point

NIM/PV Configuration Display (continued)

PVSOURCE PV Source	Auto Man Sub	Applies when PVSRCOPT = All. Field wiring or memory fetch supplies PV. PV is supplied by Operator or program. A value is substituted by a CL program.
TF PV Filter Lag Time in Minutes	_ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)

PVTV PV Target Value in EUs	_ _ _ _ _ _ _	N/A if PNTFORM = Componnt <i>range</i> = PVEULO to PVEUHI, and NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

Alarming Display

N/A if PNTFORM = Componnt

The following two example parameters represent the typical format of alarm trip points and priorities. Please refer to these as you are entering trip points and alarm priorities that follow this information. **NOTE:** If you do not enter a trip point value for a given pair of trip points and alarm priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

\$AUXUNIT Auxiliary Unit	_ _	<i>range</i> = Any configured unit in NCF <i>default</i> = - -
\$ALDLOPT Alarm Delay Option		<i>range</i> = None, OffDlyLo, OffDlyHi, OffDlyEm, OnDlyLo, OnDlyHi, OnDlyEm
\$DLYTIME Alarm delay in seconds		Applies when \$ALDLOPT is other than none.
\$REALARM (R660 and later) Re-arming Interval	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes

N/A if PNTFORM = Componnt

PVALDB PV Alarm Deadband in %	Half (of 1%) Four	One Five	Two EU	Three
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N/A if PNTFORM = Componnt, Only if PVALDB = EU

PVALDBEU PV Alarm Deadband in EU	_ _ _ _ _ _ _	<i>range</i> = ≥ 0 <i>default</i> = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLOTP to PVHHTP, and NaN <i>default</i> = NaN
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLLTP to PVHITP, and NaN <i>default</i> = NaN
BADPVPR BAD PV Alarm Priority	Emergency High Low Journal NoAction Printer JnlPrint	Reported to all Alarm Summary displays Reported to Area and Unit Alarm Summary Reported to Unit Alarm Summary display Logged but not reported to Op Stations Alarm is not reported to the system (R670 and later) Printed by printer Journaled and printed

SM Analog Input Data Point

Alarming Display

N/A if PNTFORM = Component

PVHIPR PV High Alarm Priority	NoAction (R670) Journal Printer	Low JnlPrint	High Emergency
PVLOTP PV Low Alarm Trip Point	<u> </u> <i>default</i> = NaN	<i>range</i> = PVLLTP to PVHITP, and NaN	

NIM/PV Configuration Display (continued)

PVLOPR PV Low Alarm Priority	NoAction (R670) Journal Printer	Low JnlPrint	High Emergency
BADPVPR Bad PV Alarm Priority	NoAction (R670) Journal Printer	Low JnlPrint	High Emergency

N/A if PNTFORM = Component

ALENBST Point Alarm Enable Status	Enable	<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>
	Enable	Yes	Yes	Yes
	Disable	No	Yes	Yes
	Inhibit	No	No	No

CCSRC Target Alias Address in the TRICON for the Contact Cutout Flag	<u> </u>	<i>range</i> = 0, 1 to 4000, 10001 to 14000 <i>default</i> = 0 (Not configured/UCN Access allowed for CONTCUT)
Applies only if ALMOPT = OFFNORML or CHNGOST and PNTFORM = Full		

[illegible]

SM Analog Output Data Point

Operating Configuration Display

OPTDIR (AnalogOut) AO Direct/Reverse Action	Direct Reverse	For final OP: 0% = >4 mA; 100% = >20 mA For final OP: 0% = >20 mA; 100% = >4 mA
\$OPTOL Output Tolerance in %		range: 0.0 to 106.9 default: 0.0

Mode Configuration Display

RCASOPT Remote Cascade Option	None Ddc	N/A if PNTFORM = Componnt No cascade mode of any type is allowed In Cas, AM point controls this point's OP
NMODE Analog Output Point's Normal Mode	None Man Cas	None, Man applies when RCASOPT = None. None, Man, Cas applies when RCASOPT = Ddc. No configured "normal" operating mode. Manual is the configured "normal" mode. If RCASOPT equals Ddc, Cascade may be configured as the "normal" mode.

N/A if NMODE = None, or if PNTFORM = Componnt

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Operator cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Cas Prog	Man and Cas mode, and Prog mode attribute are allowed for this data point

NAME		range =	Not applicable
Tag Name	_____	default =	Blank
NODETYPE	PM	range =	Not applicable
Node Type	APM	default =	PM
	HPM		High Performance Process Manager Module
	LM		Logic Manager Module
	SM		
SMPLTFM	TRICON		Safety Manager is using TRICON Controller
Type of Safety Control Platform	FSC		Safety Manager is using FSC Controller
PNTFORM	Full		Point is fully displayed and alarmed
Point Form	Componnt		Point is partially displayed but not alarmed
PTDESC		range =	Not applicable
Point Descriptor	_____	default =	Blank
EUDESC	_____	range =	Not applicable
Engineering Units Descriptor		default =	Blank
KEYWORD	_____	range =	Not applicable
Keyword Descriptor		default =	Blank
ASSOCDSP	_____	range =	Alphanumeric
Associated Display		default =	Blank
UNIT	____	range =	Alphanumeric
Process Unit Point is Assigned to		default =	Not applicable
NTWKNUM	____	range =	1 to 20
NIM's UCN that Contains This Point		default =	Not applicable
NODENUM	____	range =	1 to 64
SMM's Address on the UCN		default =	Not applicable
SLOTNUM	____	range =	1 to 326 for 0.5 second scan rate
Point Slot Number			1 to 652 for 1.0 second scan rate
		default =	Not applicable
The actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired.			
PRIMMOD		range =	Not applicable
Primary module Point ID	_____	default =	Null
NOSTATES	2		2 digital switch states can be configured
Number of Digital States	3		3 digital switch states can be configured
NODINPTS	0		0 digin connections can be configured
Number of Digital Input Connections	1		1 digin connection can be configured
	2		2 digin connections can be configured
NODOPTS	0		0 digout connections can be configured
Number of Digital Output Connections	1		1 digout connection can be configured
	2		2 digout connections can be configured
	3		3 digout connections can be configured

SM Digital Composite Data Point

State Configuration Display

N/A if PNTFORM = Componnt

STATETXT (1)

State 1 Descriptor

(1) |_.|

range = Not applicable

default = On

STATETXT (0)

State 0 Descriptor

(0)|_.|

range = Not applicable

default = Off

STATETXT (2)

State 2 Descriptor

(2)|_.|

range = Not applicable

default = "State2"

BOXCLR (1)

State 1 Box Color

Red
Cyan

Green (Middle box default)
Yellow

White
Blue

Black
Magenta

BOXCLR (0)

State 0 Box Color

Red
Cyan

Green
Yellow (Middle box default)

White
Blue

Black
Magenta

BOXCLR (2)

State 2 Box Color

Red (Lower box default)
Cyan

Green
Yellow

White
Blue

Black
Magenta

N/A if NODOPTS = 0

MOMSTATE

Momentary Output States

None

Mom_0

Mom_1

Mom_2

Mom_1_2

No momentary output states

State 0 is valid if (NOSTATES = 2)

State 1 is valid if (NOSTATES = 2 or 3)

State 2 is valid if (NOSTATES = 3)

State 1 or 2 is valid if (NOSTATES = 3)

N/A if PNTFORM = Componnt

LOGICSRC

Logical Input Connection Source

|_.|

DISRC(1-2)

Target Alias Addresses in the TRICON for the DI Source

Input

Source Point.Parameter

D1

|_.•.|

D2

|_.•.|

SM Digital Composite Data Point

PV/Input Configuration Display

Applies if NODINPTS = to 1		
D1_1 D1_One PV State	PVState1 PVState0	Use and display contents of STATETXT1 Use and display contents of STATETXT0
N/A if NODINPTS = 1		
D2D1_00 D2D1 Zero_Zero PV State	PVState1 PVState0 PVState2 BadPV MovPV	Use and display contents of STATETXT1 Use and display contents of STATETXT0 Use and display contents of STATETXT2 Use and display contents of BADPVTXT Use and display contents of MOVPTXT
D2D1_01 D2D1 Zero_One PV State	PVState1 PVState0 PVState2 BadPV MovPV	Use and display contents of STATETXT1 Use and display contents of STATETXT0 Use and display contents of STATETXT2 Use and display contents of BADPVTXT Use and display contents of MOVPTXT
D2D1_10 D2D1 One_Zero PV State	PVState1 PVState0 PVState2 BadPV MovPV	Use and display contents of STATETXT1 Use and display contents of STATETXT0 Use and display contents of STATETXT2 Use and display contents of BADPVTXT Use and display contents of MOVPTXT
D2D1_11 D2D1 One_One PV State	PVState1 PVState0 PVState2 MovPV BadPV	Use and display contents of STATETXT1 Use and display contents of STATETXT0 Use and display contents of STATETXT2 Use and display contents of MOVPTXT Use and display contents of BADPVTXT
N/A if PNTFORM = Componnt, or if NODINPTS = 0		
PVSRCOPT PV Source Option	OnlyAuto All	PV source selection is not available and field wiring or memory fetch supplies PV. Selection from PVSOURCE is available and Operator/CL-program may supply PV.
PVSOURCE PV Source	Auto Man Sub Track	Applies when PVSRCOPT = All. Field wiring or memory fetch supplies PV. PV is supplied by Operator or program. A value is substituted by a CL-program. PV tracks the output.

Output Configuration Display

N/A if NODOPTS < 2		
NONECONF None State Configuration	On Off	

N/A if NODOPTS = 0

default = 0 (Not Configured)

SM Digital Composite Data Point

Output Configuration Display (continued)

LMSRC	<u> </u> <u> </u> <u> </u> <u> </u> <u> </u>	range = 0, 1 to 4000, or 10001 to 14000
Target Alias Addresses in the TRICON for Local/Manual Source		default = 0 (Not Configured)

Mode Configuration Display

		N/A if NODOPTS = 0
NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Operator cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog	Man mode and Prog mode attribute are allowed for this data point
OROPT Interlock Bypass Option	Off On	
BYPASS Interlock Bypass State	Off On	Applies when OROPT = On.

Alarming Display

		N/A if PNTFORM = Componnt, or if NODINPTS = 0		
\$AUXUNIT Auxiliary Unit	_ _ _	<i>range</i> = Any configured unit in NCF <i>default</i> = - -		
\$REALARM (R660 and later) Re-alarming Interval in Minutes	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes		
ALMOPT Alarming Option	None Cmddis Offnorml	No affect Alarm if field device did not respond within configured feedback time duration. Alarm if PV and PVNORMAL differs.		
FBTIME Feedback Time in Seconds	_ _ _ _	Applies when ALMOP = Cmddis. <i>range</i> = 1 to 1000 <i>default</i> = 1000		
PVNORMAL PV Value's Normal State Text	Off	Applies when ALMOP = Offnorml. <i>range</i> = STATETXT(0) – STATETXT(2)		
ALENBST Point Alarm Enable Status		Applies when ALMOP = Offnorml, Cmddis.		
		<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>
	Enable	Yes	Yes	Yes
	Disable	No	Yes	Yes
	Inhibit	No	No	No

SM Digital Composite Data Point

Alarming Display (continued)

N/A if NODOPTS = 0

BADPVPR		Applies when ALMOPT = Offnorml, Cmddis.
BADCTLPR	NoAction (R670)	Alarm is not reported to the system.
CMDDISPR	Journal	Logged but not reported to Op Stations.
OFFNRMPR	Low	Reported to Unit Alarm Summary display
	High	Reported to Area and Unit Alarm Summary
	Emergency	Reported to all Alarm Summary displays
	Printer	Printed by printer.
	JnlPrint	Journalled and printed.
EVTOPT		Applies when ALMOPT = None, Offnorml, Cmddis.
Event Report Option	None	Neither Eip or Soe is allowed.
	Eip	Process special is triggered in AM/CG.
EIPPCODE		Applies when EVTOPT = Eip.
Event Init. Proc. Destination	_ _ _ _ _ _ _ _ _	<i>range</i> = Not applicable
		<i>default</i> = Null

CCSRC	_ _ _ _ _ _ _ _ _	Applies only if ALMOPT = Offnorml or CMDDIS and PNTFORM = Full
Target Alias Address in the TRICON for the Conact Cutout Flag		<i>range</i> = 0, 1 to 4000, 10001 to 14000
		<i>default</i> = 0 (Not Configured)

[illegible]

SM Digital Input Data Point

Operating Configuration Display

N/A if PNTFORM = Component

STATETXT (1) State 1 Description Text	(1) _ . _ . _ . _ . _ .	<i>range</i> = Not applicable <i>default</i> = "On"		
STATETXT (0) State 0 Description Text	(0) _ . _ . _ . _ . _ .	<i>range</i> = Not applicable <i>default</i> = "Off"		
BOXCLR (1) State 1 Box Color	Red Cyan	Green (Upper box default) Yellow	White Blue	Black Magenta
BOXCLR (0) State 0 Box Color	Red Cyan	Green Yellow (Lower box default)	White Blue	Black Magenta
INPTDIR Digital Input Direction		Direct Reverse	Open => pvraw = off; Closed => pvraw = on Closed => pvraw = off; Open => pvraw = on	
PVSRCOPT PV Source Option		OnlyAuto All	PV source selection is not available and field wiring or memory fetch supplies PV. Selection from PVSOURCE is available and Operator/CL-program may supply PV.	
PVSOURCE PV Source		Auto Man Sub	Applies when PVSRCOPT = All. Field wiring or memory fetch supplies PV. PV is supplied by Operator or program. A value is substituted by a CL program.	

SM Digital Input Data Point

Alarming Display

Applies only if PNTFORM = Full

\$AUXUNIT Auxiliary Unit		<i>range</i> = Any configured unit in NCF <i>default</i> = - -												
\$REALARM (R660 and later) Re-alarming Interval		Applies when ALMOPT = Offnorml. <i>range</i> = 0 to 9999 <i>default</i> = 0 minutes												
ALMOPT Alarming Option		0 = None 1 = Offnorml 3 = Chngofst												
PVNORMAL PV Value's Normal State Text		Applies when ALMOPT = Offnorml. <i>range</i> = STATETXT(0) – STATETXT(1) <i>default</i> = Text equal to "Off"												
DLYTIME Alarm Delay in Seconds		Applies when ALMOPT = Offnorml. <i>range</i> = 0 to 60 <i>default</i> = 5 (seconds)												
OFFNRMPR Off-Normal	NoAction (R670) Journal Low High Emergency Printer JnlPrint	Alarm is not reported to the system. Logged but not reported to Op Stations. Reported to Unit Alarm Summary display. Reported to Area and Unit Alarm Summary. Reported to all Alarm Summary displays. Printed by printer. Journaled and printed.												
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	Applies when ALMOPT = Offnorml. <table> <tr> <th><u>Displayed</u></th> <th><u>Logged</u></th> <th><u>Reported to EIP</u></th> </tr> <tr> <td>Yes</td> <td>Yes</td> <td>Yes</td> </tr> <tr> <td>No</td> <td>Yes</td> <td>Yes</td> </tr> <tr> <td>No</td> <td>No</td> <td>No</td> </tr> </table>	<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>	Yes	Yes	Yes	No	Yes	Yes	No	No	No
<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>												
Yes	Yes	Yes												
No	Yes	Yes												
No	No	No												
EVTOPT Event Reporting Option	None Eip EipSoe Soe	Applies only if ALMOPT = Offnorml or Chngofst. Event Initiated Processing Event Initiated Proc. and Sequence of Events Sequence of Events												
PVCHGDLY Preset in seconds for the PV Change Delay function which throttles timestamped event collection. Refer to PVCHTMTR.		N/A if EVTOPT ≠ SOE or EIP_SOE <i>range</i> = 0 to 60 <i>default</i> = 0												

SM Digital Input Data Point

Alarming Display

		Applies only if PNTFORM = Full
EIPPCODE		Applies when EVTOPT = Eip.
Event Init. Proc. Destination	_ _ _ _ _ _ _	range = Not applicable
		default = Null
		N/A if ALMOPT = None.
CCSRC		Applies only if ALMOPT = Offnorml or Chngofst.
Target Alias Address in the TRICON	_ _ _ _	range = 0, 1 to 4000, or 10001 to 14000
for the Contact Cutout Flag		default = 0 (Not Configured/UCN Access allowed for
CONTCUT)		

[illegible]

SM Digital Output Data Point

Operating Configuration Display

N/A if PNTFORM = Componnt				
STATETXT (1) State 1 Description Text	(1) _. _. _. _. _. _. _.	<i>range</i> = Not applicable <i>default</i> = "On"		
STATETXT (0) State 0 Description Text	(0) _. _. _. _. _. _. _.	<i>range</i> = Not applicable <i>default</i> = "Off"		
BOXCLR (1) State 1 Box Color	Red Cyan	Green (Upper box default) Yellow	White Blue	Black Magenta
BOXCLR (0) State 0 Box Color	Red Cyan	Green Yellow (Lower box default)	White Blue	Black Magenta
NMODATTR Normal Mode Attribute	None Operator Program		Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode	
MODEPERM Mode Permissive	Permit NotPerm		Operator can change this point's mode Operator cannot change this point's mode	
\$MODESEL (R630 and later) Mode Selection List	Man Prog		Man mode and Prog mode attribute are allowed for this data point	

SM Logic Data Point

SM Logic Data Point	Engineer:	Date:
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Point Assignment Display

NAME Tag Name		<i>range =</i> Not applicable <i>default =</i> Blank
NODETYP Node Type	PM APM HPM LM SM	<i>range =</i> Not applicable <i>default =</i> PM High Performance Process Manager Module Logic Manager Module
SMPLTFM Type of Safety Control Platform	TRICON FSC	Safety Manager is using TRICON Controller Safety Manager is using FSC Controller
PNTFORM Point Form	Full Componnt	Point is fully displayed and alarmed Point is partially displayed but not alarmed
PTDESC Point Descriptor		<i>range =</i> Not applicable <i>default =</i> Blank
ASSOCDSP Associated Display		<i>range =</i> Alphanumeric <i>default =</i> Blank
UNIT Process Unit Point is Assigned to		<i>range =</i> Alphanumeric <i>default =</i> Not applicable
NTWKNUM NIM's UCN that This Point Is In		<i>range =</i> 1 to 20 <i>default =</i> Not applicable
NODENUM SMM's Address on the UCN		<i>range =</i> 1 to 64 <i>default =</i> Not applicable
SLOTNUM Point Slot Number		<i>range =</i> 0 to 30 <i>default =</i> Not applicable
Actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired.		
PRIMMOD Primary Module ID		<i>range =</i> Not applicable <i>default =</i> Null
LOGMIX Logic Mix	12_0_12	12 inputs 0 logic blocks 12 outputs

SM Logic Data Point

NIM Input Connections Display

LISRC

Target Alias Addresses in the TRICON for the Logic Input

Input

Source Point.Parameter

L1	
L2	
L3	
L4	
L5	
L6	
L7	
L8	
L9	
L10	
L11	
L12	

LIBADOPT

Logic Bad Input Handling Option

Off

On

Hold

"Off" state is substituted for bad input.

"On" state is substituted for bad input.

Last good value is substituted for bad input.

LODSTN

Target Alias Addresses in the TRICON for the Logic output

Output

Source Point.Parameter

L1	
L2	
L3	
L4	
L5	
L6	
L7	
L8	
L9	
L10	
L11	
L12	

SM Flag Point

SM Flag Data Point	Engineer:	Date:
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Point Assignment Display

NAME			<i>range = Not applicable default = Blank</i>
Tag Name	_ . _ .		
NODETYP		PM	<i>range = Not applicable default = PM</i>
Node Type		APM	
		HPM	High Performance Process Manager Module
		LM	Logic Manager Module
		SM	
SMPLETFM		TRICON	Safety Manager is using TRICON Controller
Type of Safety Control Platform		FSC	Safety Manager is using FSC Controller
PNTFORM		Full	Point is fully displayed and alarmed
Point Form		Componnt	Point is partially displayed but not alarmed
PTDESC			<i>range = Not applicable default = Blank</i>
Point Descriptor	_ .		
EUDESC	_ .		<i>range = Not applicable default = Blank</i>
Engineering Units Descriptor			
KEYWORD	_ .		<i>range = Not applicable default = Blank</i>
Point Keyword			
ASSOCDSP	_ .		<i>range = Alphanumeric default = Blank</i>
Associated Display			
UNIT	_ .		<i>range = Alphanumeric default = Not applicable</i>
Process Unit Point is Assigned to			
NTWKNUM	_ .		<i>range = 1 to 20 default = Not applicable</i>
NIM's UCN that This Point Is In			
NODENUM	_ .		<i>range = 1 to 64 default = Not applicable</i>
SMM's Address on the UCN			
SLOTNUM	_ .		<i>range = 0 to 2000 (slots 1 to 512 are alarmable) default = Not applicable</i>
Point Slot Number			
The actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired.			
PRIMMOD	_ .		<i>range = Not applicable default = Null</i>
Primary Module ID			

Operating Configuration Display

N/A if PNTFORM = Component				
STATETXT (1) State 1 Descriptor			<i>range</i> = Not applicable <i>default</i> = ON	
STATETXT (0) State 0 Descriptor			<i>range</i> = Not applicable <i>default</i> = OFF	
BOXCLR (1) State 1 Box Color	Red Cyan	Green Yellow	White Blue	Black Magenta
BOXCLR (0) State 0 Box Color	Red Cyan	Green Yellow	White Blue	Black Magenta

SM Flag Point

Alarming Display

N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range =</i> Any configured unit in NCF <i>default =</i> - -
\$REALARM (R660 and later) Re-arming Interval in Minutes	_ _ _ _	<i>range =</i> 0 to 9999 minutes <i>default =</i> 0 minutes
OFFNRMPR	NoAction (R670) Alarm is not reported to the system. Journal Logged but not reported to Operator Stations. Low Reported to Unit Alarm Summary display. High Reported to Area and Unit Alarm Summary. Emergency Reported to all Alarm Summary displays. Printer Print only JnlPrint Journal and print	
ALENBST Alarm Enable State	Enable Disable Inhibit	<i>default =</i> Enable
EIPPCODE EIP Point ID	_ _ _ _ _ _ _	<i>range =</i> Not applicable <i>default =</i> Blank
CCSRC Contact Cutout Source	_ _ _ _	<i>range =</i> -1 to 8191 <i>default =</i> -1

SM Timer Point

SM Timer Data Point	Engineer:	Date:
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Point Assignment Display

NAME Tag Name	_____	<i>range =</i> Not applicable <i>default =</i> Blank
NODETYPE Node Type	PM APM HPM LM SM	<i>range =</i> Not applicable <i>default =</i> PM High Performance Process Manager Module Logic Manager Module
SMPLTFM Type of Safety Control Platform	TRICON FSC	Safety Manager is using TRICON Controller Safety Manager is using FSC Controller
PNTFORM Point Form	Full Componnt	Point is fully displayed and alarmed Point is partially displayed but not alarmed
PTDESC Point Descriptor	_____	<i>range =</i> Not applicable <i>default =</i> Blank
EUDESC Engineering Units Descriptor	_____	<i>range =</i> Not applicable <i>default =</i> Blank
KEYWORD Keyword Descriptor	_____	<i>range =</i> Not applicable <i>default =</i> Blank
ASSOCDSP Associated Display	_____	<i>range =</i> Alphanumeric <i>default =</i> Blank
UNIT Process Unit Point is Assigned to	_____	<i>range =</i> Alphanumeric <i>default =</i> Not applicable
NTWKNUM NIM's UCN that Contains This Point	_____	<i>range =</i> 1 to 20 <i>default =</i> Not applicable
NODENUM SMM's Address on the UCN	_____	<i>range =</i> 1 to 64 <i>default =</i> Not applicable
SLOTNUM Point Slot Number	_____	<i>range =</i> 1 to 750 for 0.5 seconds scan rate 1 to 1500 for 1.0 seconds scan rate <i>default =</i> Not applicable
The actual Range check will be calculated on the PU count for each Point type and the mix of configured Points desired.		
PRIMMOD Primary Module ID	_____	<i>range =</i> Not applicable <i>default =</i> Null
PLCADDR Target Alias Address in the TRICON for the Run/Stop Contact	_____	<i>range =</i> 0, 2001 to 4000, or 10001 to 14000 <i>default =</i> 0 (Not Configured)
PVSLTSRC (Integer) Target Alias Address in the TRICON for the Accumulator Value	_____	<i>range =</i> 0, 31001 to 31382, or 40251 to 40632 <i>default =</i> 0 (Not Configured)
SPSLTSRC Target Alias Address in the TRICON for the Preset Value	_____	<i>range =</i> 31001 to 31382, or 40251 to 40632 (Integer) <i>default =</i> 0 (Not Configured)
The Preset value is not necessarily adjacent to the Accumulator.		

SM Timer Point

NIM Operating Configuration Display

TIMEBASE Timer Base Period	SECONDS MINUTES	<i>range =</i> Seconds/Minutes <i>default =</i> Seconds
PVFORMAT PV Decimal Format	D0	D1 D2

[illegible]

SM Numeric Point

Point Assignment Display

PRIMMOD	_ _ _ _ _ _ _	<i>range</i> =	Not applicable
Primary Module ID		<i>default</i> =	Null
PVFORMAT	_ _	<i>range:</i>	D0 (XXXX.) D1 (XXX.X) D2 (XX.XX) D3 (X.XXX)
		<i>default:</i>	D0 — PVFLTOPT = Integer D2 — PVFLTOPT = Floating

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